Def. Let $f: X \to [-\infty, \infty]$ be a function where X is a topological space.
The f is said to be lower semicontinuous if:
for any $\alpha \in \mathbb{R}$, a set $\{x \mid f(x) > \alpha\} = f^{-1}[(\alpha, \infty]]$ is open set

The f is said to be upper semicontinuous if:
for any $\alpha \in \mathbb{R}$, a set $\{x \mid f(x) < \alpha\} = f^{-1}[[-\infty, \alpha)]$ is open set.

Note: since $f^{-1}[(\alpha, \infty]] \cap f^{-1}[[-\infty, \beta)] = f^{-1}[(\alpha, \beta)]$,
f is continuous if and only if f is upper and lower semicontinuous.

Prop. Let X be a topological space, and let $U \subseteq X$ be a subset.
Then, U is open set if and only if χ_U is lower semicontinuous.
and
U is closed set if and only if χ_U is upper semicontinuous.

Proof. Notice that: for any $\alpha \in \mathbb{R}$,
$\chi_U^{-1}[(\alpha, \infty]] = U$ if $\alpha < 1$ and $\chi_U^{-1}[(\alpha, \infty]] = \emptyset$ if $\alpha \geq 1$.
Therefore, χ_U is lower semicontinuous iff U is open. And,
$\chi_U^{-1}[[-\infty, \alpha)] = U^c$ if $\alpha > 0$ and $\chi_U^{-1}[[-\infty, \alpha)] = \emptyset$ if $\alpha \leq 0$
So, similarly, χ_U is upper semicontinuous iff U^c is open. ∎

Thm. Let X be a topological space.
Suppose that $\{f_i : X \to [-\infty, \infty]\}$ is a collection of lower semicontinuous,
and $\{g_i : X \to [-\infty, \infty]\}$ is a collection of upper semicontinuous.
Then, $\sup f_i$ is lower-semicontinuous
and $\inf f_i$ is upper-semicontinuous.

Proof. Let $\alpha \in \mathbb{R}$ be given. Then,

$$\left(\sup f_i\right)^{-1}[(\alpha, \infty]] = \{x \in X \mid \sup_i f_i(x) > \alpha\}$$
$$= \{x \in X \mid f_i(x) > \alpha \text{ for some } i\}$$
$$= \bigcup_i f_i^{-1}[(\alpha, \infty]]$$

So, proved. □

Def. Let $f: X \to \mathbb{C}$ be a function. Define

$$\operatorname{supp}(f) \overset{\text{def}}{=} \operatorname{cl}(\{x \mid f(x) \neq 0\}) \left(= \overline{X \setminus f^{-1}[\{0\}]}\right)$$

It is called the support of f. And, denote

$$C_c(X) \overset{\text{def}}{=} \{f \in C(X, \mathbb{C}) \mid \operatorname{supp}(f) \text{ is compact.}\}$$

Prop. $C_c(X)$ is a vector space over $\mathbb{C}$.

Proof. Let $f, g \in C_c(X)$ be given. $f+g$ is continuous clearly, and

$$\overline{\{x \mid f(x)+g(x) \neq 0\}} \subseteq \overline{\{x \mid f(x) \neq 0\} \cup \{x \mid g(x) \neq 0\}}$$

Since $\{x \mid f(x)+g(x) \neq 0\} \subseteq \{x \mid f(x) \neq 0\} \cup \{x \mid g(x) \neq 0\}$, and since $\overline{A \cup B} = \overline{A} \cup \overline{B}$, therefore $\operatorname{supp}(f+g) \subseteq \operatorname{supp}(f) \cup \operatorname{supp}(g)$.

Now, $\operatorname{supp}(f+g)$ is a closed subset in a compact set, so is compact. ∎

Def. A topological space X is called locally compact if:
given $p \in X$, there exists an open set $\mathcal{U}$ containing p s.t. $\overline{\mathcal{U}}$ is compact.

Lemma. Let X be a locally compact Hausdorff space.
If $\mathcal{U}$ is open and K is compact such that $K \subseteq \mathcal{U}$,
then there exists an open set V with compact closure such that

$$K \subseteq V \subseteq \overline{V} \subseteq \mathcal{U}$$

Proof. Since X is locally compact, given $x \in K$, there is an open $\mathcal{U}_x$ containing $\mathcal{U}_x$ such that $\overline{\mathcal{U}_x}$ is compact. Clearly, $\{\mathcal{U}_x \mid x \in K\}$ covers a compact set K, so K is covered by finitely many of these neighbors. Take $G := \bigcup_{i=1}^{n} \mathcal{U}_{x_i}$ be the such a union. Then, G is open, and

$\overline{G} = \overline{\bigcup_{i=1}^{n} \mathcal{U}_{x_i}} = \bigcup_{i=1}^{n} \overline{\mathcal{U}_{x_i}}$ is compact, being finite union of compact sets.
If $\mathcal{U} = X$, then take $V = G$. Otherwise, let $C := X \setminus \mathcal{U}$.
Then, C is a closed set such that $C \subseteq X \setminus K$.
Since X is a Hausdorff space and K is compact, given $p \in C$, there exists an open set W_p such that $K \subseteq W_p$ and $p \notin \overline{W_p}$.
Hence, $\{C \cap \overline{G} \cap \overline{W_p} \mid p \in C\}$ is a collection of compact sets with empty intersection. Since a collection of compact sets of a Hausdorff space satisfies that the finite intersection property, so for some finite points $p_1, \cdots, p_n \in C$, $C \cap \overline{G} \cap \overline{W_{p_1}} \cap \cdots \cap \overline{W_{p_n}} = \emptyset$.

So that $V := G \cap W_{p_1} \cap \cdots \cap W_{p_n}$ contains K and

$$C \cap V \subseteq C \cap \overline{V} \subseteq C \cap \left(\overline{G} \cap \overline{W_{p_1}} \cap \cdots \cap \overline{W_{p_n}}\right) = \emptyset,$$ so proved. ☐

Def. Let $f: X \to [0,1]$ be a function which has compact support.
For a compact subset K, denote $K \prec f$ if $f(x) = 1$ for all $x \in K$.
For a open set V, denote $f \prec V$ if $\mathrm{supp}(f) \subseteq V$.

Urysohn's Lemma.
Suppose that X is a locally compact Hausdorff space, and
V is open set, K is compact set such that $K \subseteq V$.
Then, there exists an $f \in C_c(X)$ such that $K \prec f \prec V$.
Proof. Define a set of dyadic rationals

$$D \overset{\text{def}}{=} \left\{ \frac{k}{2^n} \,\middle|\, n, k \in \mathbb{N},\ k < 2^n \right\} \cup \{0,1\} \subseteq [0,1] \cap \mathbb{Q}.$$

We want to show that: given $r, s \in D$ with $r < s$,
there exist open sets V_r, V_s such that $K \subseteq V_r \subseteq \overline{V_r} \subseteq V_s \subseteq \overline{V_s} \subseteq V$
and $\overline{V_r}, \overline{V_s}$ are compact.
For $n = 0$, let $0, 1 \in D$. Then, by the lemma, there exists an open V_0
such that $K \subseteq V_0 \subseteq \overline{V_0} \subseteq V$ with $\overline{V_0}$ is compact.
And, apply the lemma again for $\overline{V_0} \subseteq V$, there exists an open V_1
such that $K \subseteq V_0 \subseteq \overline{V_0} \subseteq V_1 \subseteq \overline{V_1} \subseteq V$
with $\overline{V_1}$ is compact.
Now, suppose that for $n \geq 1$, there exists a chain
$\{V_{\frac{k}{2^n}}\}_{k=0}^{2^n}$ such that $V_{\frac{k}{2^n}}$ is open set with compact closure, and

$$K \subseteq V_{\frac{0}{2^n}} \subseteq \overline{V_{\frac{0}{2^n}}} \subseteq V_{\frac{1}{2^n}} \subseteq \overline{V_{\frac{1}{2^n}}} \subseteq \cdots \subseteq V_{\frac{2^n-1}{2^n}} \subseteq \overline{V_{\frac{2^n-1}{2^n}}} \subseteq V_{\frac{2^n}{2^n}} \subseteq \overline{V_{\frac{2^n}{2^n}}} \subseteq V.$$

That is, for each $1 \leq k \leq 2^n$

$$K \subseteq V_{\frac{k-1}{2^n}} \subseteq \overline{V_{\frac{k-1}{2^n}}} \subseteq V_{\frac{k}{2^n}} \subseteq \overline{V_{\frac{k}{2^n}}} \subseteq V.$$

Proof. Page 2.

Meanwhile, by the lemma, there exists an open $V_{\frac{2k-1}{2^{n+1}}}$ such that

$$K \subseteq \overline{V_{\frac{k-1}{2^n}}} \subseteq V_{\frac{2k-1}{2^{n+1}}} \subseteq \overline{V_{\frac{2k-1}{2^{n+1}}}} \subseteq V_{\frac{k}{2^n}} \subseteq V$$

Therefore, by the induction, there is a collection of open sets

$$\{V_r \mid r \in D\}$$

satisfies that $\overline{V_r}$ is compact and $r<s$ implies $\overline{V_r} \subseteq V_s$, and $K \subseteq V_0 \subseteq \overline{V_1} \subseteq V$. Now, define for each $r, s \in D$,

$$f_r(x) = \begin{cases} 1-r & \text{if } x \in V_r \\ 0 & \text{otherwise} \end{cases}, \quad g_s(x) = \begin{cases} 1 & \text{if } x \in \overline{V_s} \\ 1-s & \text{otherwise} \end{cases}$$

and $f = \sup_{r \in D} f_r$ and $g = \inf_{s \in D} g_s$.

Clearly, $0 \le f \le 1$. And, given $x \in K$, $x \in V_r$ for every $r \in D$. So that $f(x) = \sup\{1-r \mid r \in D\} = 1$. i.e., $f|_K \equiv 1$.

And, notice that: given $x \in X$,

$$f(x) > 0 \Rightarrow f_r(x) > 0 \text{ for some } r \in D \Rightarrow x \in V_r \subseteq V_1.$$

Therefore, $\{x \in X \mid f(x) \neq 0\} \subseteq V_1$ and so that

$$\operatorname{Supp}(f) \subseteq \overline{V_1} \subseteq V.$$

Now, let's show that $f \equiv g$. Given $x \in X$, assume that $f_r(x) > g_s(x)$ for some $r, s \in D$. Then, $r<s$ and $x \in V_r$, $x \notin \overline{V_s}$. But, by the construction, $r<s \Rightarrow V_r \subseteq \overline{V_s}$, so it is not possible, so $f_r(x) \le g_s(x)$ for every $r, s \in D$, so that $f(x) \le g_s(x)$ for all s, and so $f(x) \le g(x)$. Assume that $f(x) < g(x)$ for some $x \in X$. Then, there are dyadic rationals $r', s' \in D$ such that $f(x) < 1-r' < 1-s' < g(x)$. But,

$$f(x) = \sup_{r \in D} f_r(x) < 1-r' \Rightarrow x \notin V_{r'} \text{ and } g(x) = \inf_{s \in D} g_s(x) > 1-s' \Rightarrow x \in \overline{V_{s'}}.$$

(이것도 마냥 당연한 소리는 아니다, 다냥!) But, $s' < r'$ implies $\overline{V_{s'}} \subseteq V_{r'}$, therefore $f(x) < g(x)$ cannot be happen so, $f \equiv g$. Consequently, f is lower and upper semicontinuous, so f is continuous. ∎

The Riesz Representation Theorem.

Thm. Let X be a locally compact Hausdorff space.
Let Λ be a positive linear functional on $C_c(X)$.
Then, there exists a σ-algebra $\mathfrak{M}$ in X which contains all borel sets.
and, there exists a unique positive measure μ on $\mathfrak{M}$ which represents Λ in the sence that: $\Lambda f = \int_X f \, d\mu$ for all $f \in C_c(X)$

and which has the following additional properties;

a). $\mu(K) < \infty$ for every compact set $K \subset X$.

b). For every measurable set $E \in \mathfrak{M}$,

$$\mu(E) = \inf\{\mu(V) \mid E \subseteq V, V \text{ is open}\}$$

and if $\mu(E) < \infty$, then

$$\mu(E) = \sup\{\mu(K) \mid K \subseteq E, K \text{ is compact}\}$$

c) For every open set $U \subseteq X$,

$$\mu(U) = \sup\{\mu(K) \mid K \subseteq U, K \text{ is compact}\}.$$

d). If $E \in \mathfrak{M}$, $A \subseteq E$, and $\mu(E) = 0$, then $A \in \mathfrak{M}$.